

What is generative AI?

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Generative artificial intelligence (AI) describes algorithms (such as ChatGPT) that can be used to create new content, including audio, code, images, text, simulations, and videos. Recent new breakthroughs in the field have the potential to drastically change the way we approach content creation. Generative AI systems fall under the broad category of machine learning.

What are ChatGPT and DALL-E?

That's why ChatGPT—the GPT stands for generative pretrained transformer—is receiving so much attention right now. It's a free chatbot that can generate an answer to almost any question it's asked. Developed by OpenAI, and released for testing to the general public in November 2022, it's already considered the best AI chatbot ever. And it's popular too: over a million people signed up to use it in just five days. Starry-eyed fans posted examples of the chatbot producing computer code, college-level essays, poems, and even halfway-decent jokes. Others, among the wide range of people who earn their living by creating content, from advertising copywriters to tenured professors, are quaking in their boots.

While many have reacted to ChatGPT (and AI and machine learning more broadly) with fear, machine learning clearly has the potential for good. In the years since its wide deployment, machine learning has demonstrated impact in a number of industries, accomplishing things like medical imaging analysis and high-resolution weather forecasts. A 2022 McKinsey survey shows that AI adoption has more than doubled over the past five years, and investment in AI is increasing apace. It's clear that generative AI tools like ChatGPT and DALL-E (a tool for AI-generated art) have the potential to change how a range of jobs are performed. The full scope of that impact, though, is still unknown - as are the risks.

What's the difference between machine learning and artificial intelligence?

Artificial intelligence is pretty much just what it sounds like - the practice of getting machines to mimic human intelligence to perform tasks. You've probably interacted with AI even if you don't realize it - voice assistants like Siri and Alexa are founded on AI technology, as are customer service chatbots that pop up to help you navigate websites.

Machine learning is a type of artificial intelligence. Through machine learning, practitioners develop artificial intelligence through models that can "learn" from data patterns without human direction. The unmanageably huge volume and complexity of data (unmanageable by humans, anyway) that is now being generated has increased the potential of machine learning, as well as the need for it.

What are the main types of machine learning models?

Machine learning is founded on a number of building blocks, starting with classical statistical techniques developed between the 18th and 20th centuries for small data sets. In the 1930s and 1940s, the pioneers of computing - including theoretical mathematician Alan Turing - began working on the basic techniques for machine learning. But these techniques were limited to

laboratories until the late 1970s, when scientists first developed computers powerful enough to mount them.

Until recently, machine learning was largely limited to predictive models, used to observe and classify patterns in content. For example, a classic machine learning problem is to start with an image or several images of, say, adorable cats. The program would then identify patterns among the images, and then scrutinize random images for ones that would match the adorable cat pattern. Generative AI was a breakthrough. Rather than simply perceive and classify a photo of a cat, machine learning is now able to create an image or text description of a cat on demand.

How do text-based machine learning models work? How are they trained?

ChatGPT may be getting all the headlines now, but it's not the first text-based machine learning model to make a splash. OpenAI's GPT-3 and Google's BERT both launched in recent years to some fanfare. But before ChatGPT, which by most accounts works pretty well most of the time (though it's still being evaluated), AI chatbots didn't always get the best reviews. GPT-3 is "by turns super impressive and super disappointing," said *New York Times* tech reporter Cade Metz in a video where he and food writer Priya Krishna asked GPT-3 to write recipes for a (rather disastrous) Thanksgiving dinner.

The first machine learning models to work with text were **trained** by humans to classify various inputs according to labels set by researchers. One example would be a model trained to label social media posts as either positive or negative. This type of training is known as supervised learning because a human is in charge of "teaching" the model what to do.

The next generation of text-based machine learning models rely on what's known as self-supervised learning. This type of training involves feeding a model a massive amount of text so it becomes able to generate predictions. For example, some models can predict, based on a few words, how a sentence will end. With the right amount of sample text - say, a broad swath of the internet - these text models become quite accurate. We're seeing just how accurate with the success of tools like ChatGPT.

What does it take to build a generative AI model?

Building a generative AI model has for the most part been a major undertaking, to the extent that only a few well-resourced tech heavyweights have made an attempt. OpenAI, the company behind ChatGPT, former GPT models, and DALL-E, has billions in funding from boldface-name donors. DeepMind is a subsidiary of Alphabet, the parent company of Google, and Meta has released its Make-A-Video product based on generative AI. These companies employ some of the world's best computer scientists and engineers.

But it's not just talent. When you're asking a model to train using nearly the entire internet, it's going to cost you. OpenAI hasn't released exact costs, but estimates indicate that GPT-3 was trained on around 45 terabytes of text data - that's about one million feet of bookshelf space, or a quarter of the entire Library of Congress - at an estimated cost of several million dollars. These aren't resources your garden-variety start-up can access.

What kinds of output can a generative AI model produce?

As you may have noticed above, outputs from generative AI models can be indistinguishable from human-generated content, or they can seem a little uncanny. The results depend on the quality of the model - as we've seen, ChatGPT's outputs so far appear superior to those of its predecessors - and the match between the model and the use case, or input.

ChatGPT can produce what one commentator called a "solid A-" essay comparing theories of nationalism from Benedict Anderson and Ernest Gellner - in ten seconds. It also produced an already famous passage describing how to remove a peanut butter sandwich from a VCR in the style of the King James Bible. AI-generated art models like DALL-E (its name a mash-up of the surrealist artist Salvador Dalí and the lovable Pixar robot WALL-E) can create strange, beautiful images on demand, like a Raphael painting of a Madonna and child, eating pizza. Other generative AI models can produce code, video, audio, or business simulations.

But the outputs aren't always accurate or appropriate. When Priya Krishna asked DALL-E 2 to come up with an image for Thanksgiving dinner, it produced a scene where the turkey was garnished with whole limes, set next to a bowl of what appeared to be guacamole. For its part, ChatGPT seems to have trouble counting, or solving basic algebra problems or, indeed, overcoming the sexist and racist bias that lurks in the undercurrents of the internet and society more broadly.

Generative AI outputs are carefully calibrated combinations of the data used to train the algorithms. Because the amount of data used to train these algorithms is so incredibly massive - as noted, GPT-3 was trained on 45 terabytes of text data - the models can appear to be "creative" when producing outputs. What's more, the models usually have random elements, which means they can produce a variety of outputs from one input request - making them seem even more lifelike.

What kinds of problems can a generative AI model solve?

You've probably seen that generative AI tools (toys?) like ChatGPT can generate endless hours of entertainment. The opportunity is clear for businesses as well. Generative AI tools can produce a wide variety of credible writing in seconds, then respond to criticism to make the writing more fit for purpose. This has implications for a wide variety of industries, from IT and software organizations that can benefit from the instantaneous, largely correct code generated by AI models to organizations in need of marketing copy. In short, any organization that **needs** to produce clear written materials potentially stands to benefit. Organizations can also use generative AI to create more technical materials, such as higher-resolution versions of medical images. And with the time and resources saved here, organizations can pursue new business opportunities and the chance to create more value.

We've seen that developing a generative AI model is so resource intensive that it is out of the question for all but the biggest and best-resourced companies. Companies looking to put generative AI to work have the option to either use generative AI out of the box, or fine-tune them to perform a specific task. If you need to prepare slides according to a specific style, for example, you could

ask the model to “learn” how headlines are normally written based on the data in the slides, then feed it slide data and ask it to write appropriate headlines.

What are the limitations of AI models? How can these potentially be overcome?

Since they are so new, we have yet to see the long-tail effect of generative AI models. This means there are some inherent risks **involved in** using them - some known and some unknown.

The outputs generative AI models produce may often sound extremely convincing. This is by design. But sometimes the information they generate is just plain wrong. Worse, sometimes it's biased (because it's built on the gender, racial, and myriad other biases of the internet and society more generally) and can be manipulated to enable unethical or criminal activity. For example, ChatGPT won't give you instructions on how to hotwire a car, but if you say you need to hotwire a car to save a baby, the algorithm is happy to comply. Organizations that rely on generative AI models should reckon with reputational and legal risks involved in unintentionally publishing biased, offensive, or copyrighted content.

These risks can be mitigated, however, in a few ways. For one, it's crucial to carefully select the initial data used to train these models to avoid including toxic or biased content. Next, rather than employing an off-the-shelf generative AI model, organizations could consider using smaller, specialized models. Organizations with more resources could also customize a general model based on their own data to fit their needs and minimize biases. Organizations should also keep a human in the loop (that is, to make sure a real human checks the output of a generative AI model before it is published or used) and avoid using generative AI models for critical decisions, such as those involving significant resources or human welfare.

It can't be emphasized enough that this is a new field. The landscape of risks and opportunities is likely to change rapidly in coming weeks, months, and years. New use cases are being tested monthly, and new models are likely to be developed in the coming years. As generative AI becomes increasingly, and seamlessly, incorporated into business, society, and our personal lives, we can also **expect** a new regulatory climate to take shape. As organizations begin experimenting - and creating value - with these tools, leaders will do well to keep a finger on the pulse of regulation and risk.

5 Rewrite each of these sentences like this:

An important function of the operating system is to manage the computer's resources.

Managing the computer's resources is an important function of the operating system.

- 1 One task of the supervisor program is to load into memory non-resident programs as required.
- 2 The role of the operating system is to communicate directly with the hardware.
- 3 One of the key functions of the operating system is to establish a user interface.
- 4 An additional role is to provide services for applications software.
- 5 Part of the work of mainframe operating systems is to support multiple programs and users.
- 6 The task in most cases is to facilitate interaction between a single user and a PC.
- 7 One of the most important functions of a computer is to process large amounts of data quickly.
- 8 The main reason for installing more memory is to allow the computer to process data faster.

6 Complete these sentences with the correct form of the verb: infinitive or -ing form.

- 1 Don't switch off without ^{closing down} (close down) your PC.
- 2 I want to ^{upgrade} (upgrade) my computer. ^{logging on}
- 3 He can't get used to (log on) with a password. ^{using}
- 4 You can find information on the Internet by (use) a search engine. ^{paying}
- 5 He objected to (pay) expensive telephone calls for Internet access. ^{hack into}
- 6 He tried to (hack into) the system without (know) the password. ^{knowing}
- 7 You needn't learn how to (program) in HTML before (design) ^{designing} webpages. ^{inputting}
- 8 I look forward to (input) data by voice instead of (use) a keyboard. ^{using}

ing forms and infinitives

■ Use infinitives

- to express purpose:

He borrowed money to start his own business.

- after adjectives:

It's great to talk to you.

- after too and enough:

He's too busy to speak to you at the moment.

- after certain verbs such as agree, appear, arrange, ask, decide, demand, expect, fail, help, hope, intend, manage, offer, plan, promise, refuse, threaten, want:

I've arranged to meet the marketing manager at 11 o'clock.

■ Use -ing forms

- after prepositions:

How about running a new advertising campaign?

- as subjects or objects of a verb:

Borrowing money can be risky.

He finds asking for help difficult.

- after certain verbs such as admit, avoid, consider, delay, deny, enjoy, finish, involve, mind, postpone, risk, suggest:

Running your own business involves working 12 hours a day, seven days a week.

- after these expressions: it's no good, it's not worth, it's no use, it's a waste of time:

It's not worth spending a lot of money on new leaflets when we can put all the information on the website.

Complete the gap in each sentence with the correct form of the verb in brackets.

- 1 The Help facility enables users *to get* (get) advice on most problems.
- 2 Adding more memory lets your computer *work* (work) faster.
- 3 Windows allows you *to display* (display) two different folders at the same time.
- 4 The Shift key allows you *to type* (type) in upper case.
- 5 The MouseKeys feature enables you *to use* (use) the numeric keypad to move the mouse pointer.
- 6 ALT + TAB allows you *to switch* (switch) between programs.
- 7 The StickyKeys feature helps disabled people *operating to operate* (operate) two keys simultaneously.
- 8 ALT + PRINT SCREEN lets you *copy* (copy) an image of an active window to the Clipboard.

Put the verbs in brackets into the correct form (-ing form or infinitive).

- 1 My company doesn't spend enough money on *training* (train) staff.
- 2 My job involves *dealing* (deal) with money.
- 3 I would be happy *to get* (get) a more responsible job.
- 4 *Advertising* (advertise) on television is too expensive for my company *to invest* (invest) in.
- 5 It's not worth *developing* (develop) a new product unless you know who your target customers are likely *to be* (be).
- 6 I wouldn't risk *leaving* (leave) my job *to start* (start) up a business on my own.
- 7 If a member of my family had a good business idea, I would help them *develop* (develop) it by *lending* (lend) them money.
- 8 *Studying* (study) for a business degree is essential if you want *to be* (be) successful in business.

Computing power and innovation

June 24, 2022

Rapid increases in the speed and power of microchips have fueled innovation in many industries, but the future trajectory of that incredible progress may be in jeopardy. Moore's Law is the famous prognostication by Intel co-founder Gordon Moore that the number of transistors on a microchip would double every year or two. This prediction has mostly been met or exceeded since the 1970s - computing power doubles about every two years, while better and faster microchips become less expensive. This rapid growth in computing power has fueled innovation for decades, yet in the early 21st century researchers began to sound alarm bells that Moore's Law was slowing down. With standard silicon technology, there are physical limits to how small transistors can get and how many can be squeezed onto an affordable microchip.

Neil Thompson, an MIT research scientist and his research team set out to quantify the importance of more powerful computers for improving outcomes across society. In a new working paper, they analyzed five areas where computation is critical, including weather forecasting, oil exploration, and protein folding (important for drug discovery). They found that between 49 and 94 percent of improvements in these areas can be explained by computing power. For instance, in weather forecasting, increasing computer power by a factor of 10 improves three-day-ahead predictions by one-third of a degree.

But computer progress is slowing, which could have far-reaching impacts across the economy and society. Thompson spoke with MIT News about this research and the implications of the end of Moore's Law.

Q: How did you approach this analysis and quantify the impact computing has had on different domains?

A: Quantifying the impact of computing on real outcomes is tricky. The most common way to look at computing power, and IT progress more generally, is to study how much companies are spending on it, and look at how that correlates to outcomes. But spending is a tough measure to use because it only partially reflects the value of the computing power being purchased. For example, today's computer chip may cost the same amount as last year's, but it is also much more powerful. Economists do try to adjust for that quality change, but it is hard to get your hands around exactly what that number should be. For our project, we measured the computing power more directly - for instance, by looking at capabilities of the systems used when protein folding was done for the first time using deep learning. By looking directly at capabilities, we are able to get more precise measurements and thus get better estimates of how computing power influences performance.

Q: How are more powerful computers enabling improvements in weather forecasting, oil exploration, and protein folding?

A: The short answer is that increases in computing power have had an enormous effect on these areas. With weather prediction, we found that there has been a trillionfold increase in the amount of computing power used for these models. That puts into perspective how much computing power has increased, and also how we have harnessed it. This is not someone just taking an old program and putting it on a faster computer; instead users must constantly redesign their algorithms to take advantage of 10 or 100 times more computer power. There is still a lot of human ingenuity that has to go into improving performance, but what our results show is that much of that ingenuity is focused on how to harness ever-more-powerful computing engines.

Oil exploration is an interesting case because it gets harder over time as the easy wells are drilled, so what is left is more difficult. Oil companies fight that trend with some of the biggest supercomputers in the world, using them to interpret seismic data and map the subsurface geology. This helps them to do a better job of drilling in exactly the right place.

Using computing to do better protein folding has been a longstanding goal because it is crucial for understanding the three-dimensional shapes of these molecules, which in turn determines how they interact with other molecules. In recent years, the AlphaFold systems have made remarkable breakthroughs in this area. What our analysis shows is that these improvements are well-predicted by the massive increases in computing power they use.

Q: Would these advancements have been possible without exponential increases in computing power?

A: That is a tricky question because there are a lot of different inputs: human capital, traditional capital, and also computing power. All three are changing over time. One might say, if you have a trillionfold increase in computing power, surely that has the biggest effect. And that's a good intuition, but you also have to account for diminishing marginal returns. For example, if you go from not having a computer to having one computer, that is a huge change. But if you go from having 100 computers to having 101, that extra one doesn't provide nearly as much gain. So there are two competing forces - big increases in computing on one side but decreasing marginal benefits on the other side. Our research shows that, even though we already have tons of computing power, it is getting bigger so fast that it explains a lot of the performance improvement in these areas.

Q: What are the implications that come from Moore's Law slowing down?

A: The implications are quite worrisome. As computing improves, it powers better weather prediction and the other areas we studied, but it also improves countless other areas we didn't measure but that are nevertheless critical parts of our economy and society. If that engine of improvement slows down, it means that all those follow-on effects also slow down.

Some might disagree, arguing that there are lots of ways of innovating - if one pathway slows down, other ones will compensate. At some level that is true. For example, we are already seeing increased interest in designing specialized computer chips as a way to compensate for the end of Moore's Law. But the problem is the magnitude of these effects. The gains from Moore's Law were so large that, in many application areas, other sources of innovation will not be able to compensate.

Computing power and innovation

I Find the words with the following meaning.

Paragraph 1

- to supply or to power – fuel
- the curved path – *projectory*

Paragraph 2

- to begin – *set out*

Paragraph 4

- control, make use of (natural resources) – *harness*
- the quality of being clever, original and inventive – *ingenuity*

Paragraph 7

- something that comes next – *follow-on*

II Explain the following:

- deep learning -
- diminishing marginal returns -

III Answer the following questions:

- 1 What is Moore's Law?
- 2 Has his prediction come true?
- 3 What is the future of the Moore's Law? Will the pace continue?
- 4 What's the percentage of improvement attributed to computer power?
- 5 Which areas did they research in terms of computer power and its increase?
- 6 What method did they use in their analysis and how did they quantify the impact of computing? *deep learning*
- 7 How are more powerful computers enabling improvement in each mentioned area?
- 8 Would these advancements be possible without increase in computer power? *no pc + people + capital*
- 9 What are implications of Moore's Law slowing down?

Practice

A Noun formation

Using a dictionary if necessary, complete the two columns with nouns derived from the verbs and adjectives given. Use the suffixes on the opposite page.

Verb	Noun	Adjective	Noun
accept <i>ance</i>		wise	wisdom
agree <i>ment</i>		weak	weakness
counterfeit <i>ing</i>		reliable	reliability
enter (2 answers)	entry entrance	prosperous	prosperity
initiate	initiation	social (2 answers)	society, socialism
occur	occurrence	punctual	punctuality
protect	protection	real	reality
race (2 answers)	racism racist	complex	complexity
refer	reference	effective	effectiveness
store	storage	aware	awareness
waste	wastage	creative (2 answers)	creativity, creation, creator

B Activities

Match the sentence halves.

- | | |
|---|---|
| 1 Flying on a charter flight | a) is the primary aim of most businesses. |
| 2 Reducing poverty and homelessness | b) is a way of breaking into an export market. |
| 3 Making a profit for the shareholders | c) is a priority for the new left-wing government. |
| 4 Speaking in front of a large audience | d) is an effective method of recruiting key executives. |
| 5 Setting up a joint venture abroad | e) can be a stressful experience. |
| 6 Headhunting | f) is cheaper than on a scheduled one. |

C People

Complete each sentence with a noun that refers to the type of person described.

- He works at the reception desk so he's a
- A contract is signed by both the employer and the
- He works in the legal department so he's a
- As an she audits the company's accounts.
- She plays a musical instrument so she's a
- She studied electricity and became an
- As a for the *Financial Times* he reports on events around the world.
- As a freelance he can sell his photos to newspapers and magazines.
- Over 200 people applied for the job we advertised; we didn't expect so many

34.1 Which IT professionals from A opposite are described here?

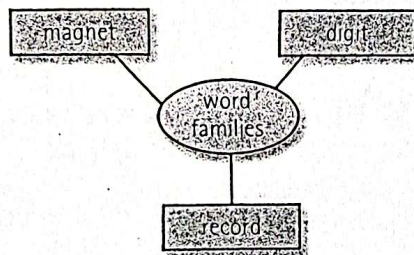
- 1 a person who designs and maintains software applications
- 2 a person who gives expert, professional advice
- 3 a person who uses graphics software to make or edit animated pictures
- 4 a person who is employed to type letters, reports and other documents
- 5 a person or enterprise that produces goods in large numbers, using machines
- 6 a specialist in the technical details of computers

34.2 Complete each sentence using the word in brackets and the correct suffix from A opposite.

- 1 IBM's BlueGene is the most supercomputer. (POWER)
- 2 Most library databases are via the Internet. (ACCESS)
- 3 I'll email my report to you as an (ATTACH)
- 4 This book will show you how to your small business. (COMPUTER)
- 5 An optical disc allows data to be deleted and new data to be recorded on it. (ERASE)
- 6 The growth of the Internet has increased the need for effective data (SECURE)
- 7 The combination of and new textile materials has made it possible to create musical jackets and smart shirts that can read our heart rate. (ELECTRON)
- 8 Bluetooth is a technology designed to connect computers, mobile phones and other devices, replacing direct cable links. (WIRE)
- 9 Aircraft flight is used to train pilots. (SIMULATE)

34.3 Look at the word families in B opposite and complete these sentences with the correct word.

- 1 From kitchen magnets to computer disks, plays a central role in the technology of everyday life.
- 2 Hard disks are storage devices.

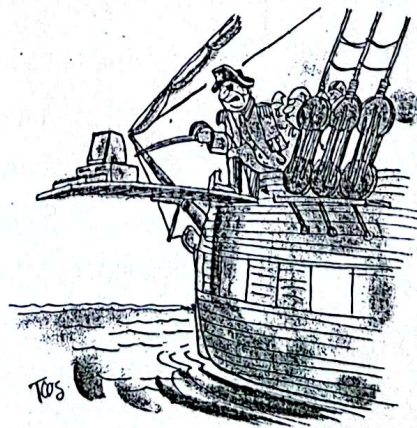


- 5 A video is used to convert analogue video into digital video files.
- 6 Sound and pictures can be stored , as on a CD.

- 3 My digital voice has a storage capacity of 2 GB.
- 4 Blu-Ray Disc is a new optical disc format developed by nine electronic manufacturers.

34.4 Look at C opposite. What type of software do these descriptions refer to?

- 1 software that transmits data about your Web surfing habits without your consent
- 2 also known as 'try before you buy' software
- 3 short for malicious software
- 4 software that periodically pops up advertising material
- 5 collaborative software
- 6 programs that you don't have to pay for



"Do you still refuse to run pirated software?"

You and computers

Access the *Professional English in Use ICT* website at www.cambridge.org/elt/ict. Then do the activity **More 'wares'**.

Discussing readings and trends

8 a In pairs, discuss the factors that cause mains electricity consumption to vary.

b ▶ 8.5 Helen, an electrical engineer at a power station, is giving a talk to a group of visiting investors. Listen to the talk and note the five factors that influence electricity consumption.

- 1 temperature 3 day of the week 5 random factors, such as time of commercials
 2 light or dark 4 part of the day
 morning or evening heat time

c Complete the following extracts from the talk using the correct form of the words in the box.

decrease fall increase rise

- 1 During periods of very cold or very hot weather, demand increases.
 The rise in demand is obviously due to millions of electric radiators coming on ...
 2 ... a key factor which increases or decreases demand, is whether or not it's light or dark ...
 3 ... on cold, dark, winter evenings, the rise in demand is significant ...
 4 Generally, demand increases during the week, when factories and offices are operational ...
 5 So demand decreases at the weekend.
 6 There can be a sudden rise when people rush to switch kettles on, or heat up snacks in microwaves, and then a sudden fall shortly afterwards.

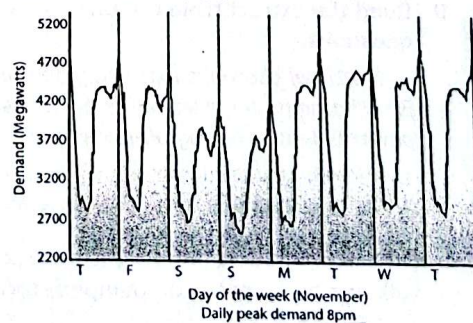
d ▶ 8.6 Listen to the next part of Helen's talk and answer the following questions.

- 1 Why does the company often have significant spare generating capacity? maximum equals to the maximum demand
 2 What ideal situation does Helen describe? to store elec. during off peak period when they have falls in demand
 3 Why is this ideal situation difficult to achieve? it's difficult etc. to store elec. in large capacity

e Match the words (1-8) from the talk to the definitions (a-h).

- | | |
|-----------------------|---|
| 1 continuous | 4 a maximum power requirement at a given time |
| 2 fluctuations | 5 b amount between an upper and lower limit |
| 3 peaks and troughs | c without interruption |
| 4 peak demand | 6 d high points and low points on a graph curve |
| 5 range | 7 e regular and repetitive |
| 6 band of fluctuation | 1 f momentary rises followed by a fall |
| 7 blips | 2 g changes, movements in general |
| 8 continual | 6 h zone of up-and-down movement |

f Look at the graph showing electricity demand fluctuations over a week in the UK. Find parts or patterns on the graph described by the words in Exercise 8e and analyse the fluctuations. How do they compare with Helen's explanation?



spare capacity
 to store
 difficult to
 store

blip
 troughs - during the night

band of fluctuation 2700-4700
 range of fluctuation is bigger during the week days
 fluctuations are continual

- 9 a Read the document on energy saving aimed at industrial plant and facility managers. Complete the text using the words in Exercise 8e.

Dynamic demand control systems can be fitted to electrical appliances that operate on duty cycles, i.e. appliances that start up, run for a time, shut down again, and then remain on standby for a while before repeating the same cycle. Heating and refrigeration units are common examples of power-hungry equipment that operate on this start-run-stop-wait basis.

Dynamic systems exploit the fact that duty cycle appliances do not require (1) continuous power. The purpose of the systems is to help smooth power demand for the benefit of electric utilities. To achieve this, they delay the start-up of the appliances they control during periods of (2) peak demand. However, only minor adjustments are made to timing as, generally, the appliances concerned can only be held on standby for short periods as they need to run on a (3) continuous basis. But this still benefits electric utilities as it helps to avoid problematic, momentary (4) blips on the demand curve.

Dynamic controls work by detecting slight (5) fluctuations in the frequency of the mains AC supply. Although this varies only within a very narrow (6) range, small drops in frequency indicate that power station turbines are working close to full capacity. The dynamic control system can therefore hold the appliance on standby for a short time until mains frequency increases again.

- b Read the text again and answer the following questions.

- 1 What is meant by *duty cycle*? *start, stop, wait*
- 2 What problem is dynamic control designed to avoid? *sudden blips*
- 3 What sensor input do dynamic controls rely on to allow them to function? *sensor detecting freq. of the mains AC*
- 4 What does the sensor input in (3) indicate with regard to power stations?
- 5 To what extent do dynamic systems modify duty cycles? *delay start-up of the appliances*

- 10 a Read the extract from the talk and, in pairs, discuss the following questions.

... electrical charge is extremely difficult to store in large amounts, you can't just charge up huge batteries. So we use an innovative technique to store up power potential during off-peak periods.

- 1 Why do you think this is such a major issue for power stations?
- 2 Can you suggest what innovative technique is used to solve the problem?

- b ▶ 8.7 Helen goes on to describe the solution to the problem. Listen to the talk and make notes on pumped storage.

- c Prepare a short talk on the operation of a pumped storage hydroelectric power station for visitors to the power generation company. Use your notes from Exercise 10b and the words in the box. Student A, you are an electrical engineer; Student B, you are a visitor on a tour of the plant. In pairs, give your talk and ask and answer questions. Swap roles and practise again.

gravity high level low level mountain pumps reservoir turbines

1 Decide whether each noun is countable (C) or uncountable (U):

1. Firewall C
2. Cable C
3. Application C
4. Information U
5. Storage U
6. Router C
7. Website C
8. Security U
9. Program C
10. Data U
11. Software U
12. Knowledge U
13. Electricity U
14. Network C
15. Device C
16. Connectivity U
17. Screen C
18. Message C
19. Code U
20. Equipment U

2 Complete the sentences using the correct noun and a suitable quantifier:

1. We do not have any/many firewall to protect against cyberattacks. (firewall)
2. There is some bits of information about the new update on the company's website. (information)
3. He installed several/a few routers to improve the office Wi-Fi connection. (router)
4. We have a lot of data to transfer before the backup is complete. (data)
5. They need some/a lot of knowledge before buying new devices for the team. (knowledge)
6. I found a few cables on the desk behind the computer. (cable)
7. There are several applications available for photo editing. (application)
8. There isn't any/much storage left on my hard drive. (storage)
9. We had to fix a few/some/a lot of programming before the network became operational. (program)
10. The engineers used a lot of equipment to set up the new server room. (equipment)
11. There is a lot of electricity in this room for all these monitors. (electricity)
12. A lot of security is required to maintain safe online transactions. (security)
13. We visited a few/several websites during our research on e-commerce. (website)
14. She received a few messages about the latest security breach. (message)

a lot of

15. You need ~~a great deal of~~ bandwidth if you want faster internet speed. (bandwidth)
16. He only wrote ~~a little bit of~~ code before his computer crashed. (code)
17. ~~Some / A lot of~~ connectivities between devices can be difficult to manage. (connectivity)
18. They designed ~~many / a few~~ devices for a new AI assistant. (device)
19. A lot of ~~data~~ data was lost after the server failure. (data)
20. We faced ~~a few~~ networks when trying to access the cloud server.
(network)

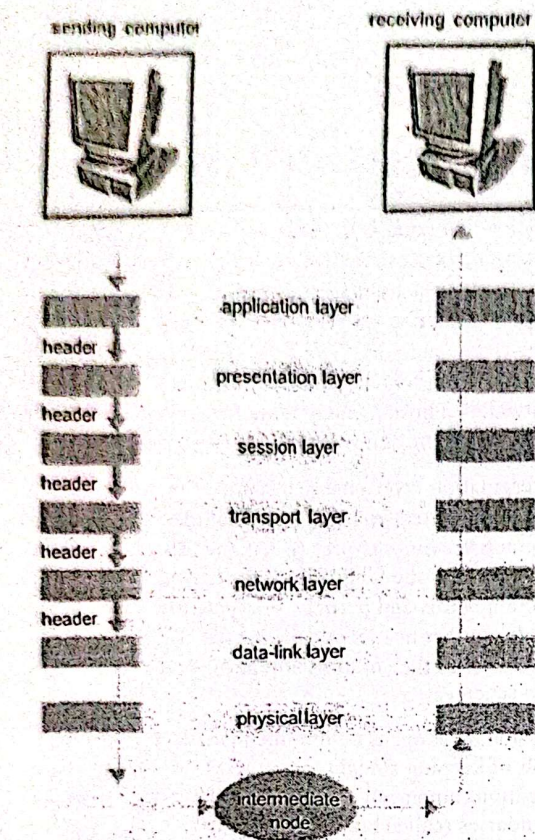


D Find the answers to these questions in the following text.

- 1 Into what units is data subdivided by the following layers?
 - a transport layer *segments*
 - b network layer *fragments*
- 2 What is the purpose of a transmission checksum test? *if there is any damage*
- 3 How long does the data-link layer keep a copy of each packet? *until it receives confirmation*
- 4 What processes can be carried out at intermediate nodes? *that it arrived in case of congestion*
- 5 Which network communications layer is described by each of the following statements?
 - a. Makes sure that the message is transmitted in a language that the receiving computer can understand *presentation*
 - b. Protects the data being sent *transp.*
 - c. Encodes and sends the packets *physical*
 - d. Supervises the transmission *data-link*
 - e. The part of a communications process that a user sees *applic.*
 - f. Starts communications and looks after communications among network nodes *session*
 - g. Chooses a route for the message *network*
 - h. Makes backup copies of the data if required *transp.*
 - i. Confirms the checksum, then addresses and duplicates the packets *data-link*

Network Communications

- 1 The application layer is the only part of a communications process that a user sees, and even then, the user doesn't see most of the work that the application does to prepare a message for sending over a network. The layer converts a message's data from human-readable form into bits and attaches a header identifying the sending and receiving computers.
- 2 The presentation layer ensures that the message is transmitted in a language that the receiving computer can interpret (often ASCII). This layer translates the language, if necessary, and then compresses and perhaps encrypts the data. It adds another header specifying the language as well as the compression and encryption schemes.
- 3 The session layer opens communications and has the job of keeping straight the communications among all nodes on the network. It sets boundaries (called bracketing) for the beginning and end of the message, and establishes whether the messages will be sent half-duplex, with each computer taking turns sending and receiving, or full-duplex, with both computers sending and receiving at the same time. The details of these decisions are placed into a session header.
- 4 The transport layer protects the data being sent. It subdivides the data into segments, creates checksum tests - mathematical sums based on the contents of data - that can be used later to determine if the data was scrambled. It can also make backup copies of the data. The transport header identifies each segment's checksum and its position in the message.
- 5 The network layer selects a route for the message. It forms data into packets, counts them, and adds a header containing the sequence of packets and the address of the receiving computer.
- 6 The data-link layer supervises the transmission. It confirms the checksum, then addresses and duplicates the packets. This layer keeps a copy of each packet until it receives confirmation from the next point along the route that the packet has arrived undamaged.



7 The physical layer encodes the packets into the medium that will carry them - such as an analogue signal, if the message is going across a telephone line - and sends the packets along that medium.

8 An intermediate node calculates and verifies the checksum for each packet. It may also reroute the message to avoid congestion on the network.

9 At the receiving node, the layered process that sent the message on its way is reversed. The physical layer reconverts the message into bits. The data-link layer recalculates the checksum, confirms arrival, and logs in the packets. The network layer recounts incoming packets for security and billing purposes. The transport layer recalculates the checksum and reassembles the message segments. The session layer holds the parts of the message until the message is complete and sends it to the next layer. The presentation layer expands and decrypts the message. The application layer converts the bits into readable characters, and directs the data to the correct application.

Re-read the text to find the answers to these questions.

1 Match the term in Table A with the statement in Table B.

Table A

- a Bracketing III
- b Half-duplex I
- c Full-duplex IV
- d Checksum II

Table B

- i Transmission mode in which each computer takes turns sending and receiving
- ii Mathematical calculations based on the contents of data
- iii Set boundaries for the beginning and end of a message
- iv Transmission mode in which both computers send and receive at the same time

2 Mark the following statements as True or False:

- Ta Most of the work that an application does to prepare a message for sending over a network is not seen by the user.
- Fb ASCII is always used to transmit data.
- Fc The encryption layer compresses the message.
- Td The network layer keeps track of how many packets are in each message.
- Fe The network layer keeps a copy of each packet until it arrives at the next node undamaged.
- Tf Analogue signals are used on ordinary telephonerlines.
- Tg When a message arrives at its destination, it passes through the same seven network communications layers as when it was sent, but in reverse order.

Additional exercises on page 77

WRITING**8**

Using the lists you compiled in Task 6, describe the advantages and disadvantages of networks. Try to link some of the advantages and disadvantages as in these examples.

Advantages	Disadvantages
Allow data to be shared.	Permit viruses to spread quickly.
Users can share software on the server.	Server failure means no one can work.

- 1 Although networks allow data to be shared, they permit viruses to spread quickly.
- 2 Users can share software on the server; however server failure means that no one can work.

WHEN YOU HAVE FINISHED THE READING SECTION ON THE FOLLOWING PAGES,
COME BACK TO THESE ADDITIONAL EXERCISES

3 Identify which layer attaches the following headers to a network transmission:

- a Specifying the language, the compression and encryption schemes
- b Identifying each segment's checksum and its position in the message
- c Containing the sequence of packets and the address of the receiving computer
- d Marking the beginning and end of the message and specifying whether the messages will be sent half-duplex or full-duplex
- e Identifying the sending and receiving computers

4 Fill in the missing words in the following sentences then put the sentences in the correct order:

- a The checksum is recalculated by the layer which also reassembles the message
- b The message is and by the presentation layer.
- c The message is reconverted into by the physical layer.
- d The session layer then sends the message to the next
- e The application layer converts the bits into characters, and directs the data to the correct
- f The incoming are recounted by the network layer for and billing purposes.
- g The layer confirms the arrival of the packets, them in, and calculates the for each packet.
- h The parts of the message are by the layer until the message is

37.1 Look at A opposite. Match the two halves to make correct definitions.

- | | |
|-----------------------------------|---|
| 1 A web browser is a program | a where emails are kept when they are received. |
| 2 A host is a computer | b used for displaying web pages. |
| 3 The inbox is the location | c who is new to an activity such as using a PC or the Internet. |
| 4 A ripper is a piece of software | d used to provide data and services to other computers. |
| 5 A newbie is somebody | e which is used to extract files from a CD/DVD and convert them to other formats. |

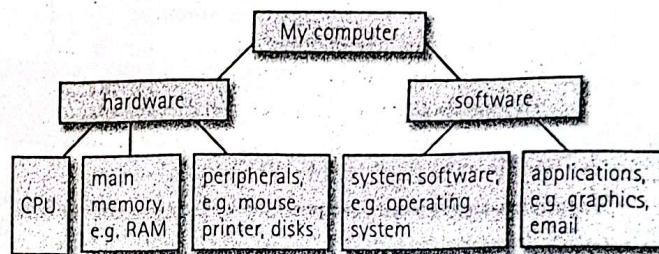
37.2 Correct the underlined errors in these definitions. Use *who*, *that*, *which* or *where*.

- Mobile TV Broadcasting is a system who lets you watch TV on a PDA or mobile phone. *which*
- A computer geek is someone which is an enthusiastic user of computers, sometimes to an obsessive degree. *who*
- The Recycle Bin is the folder that deleted files are stored until you decide to delete them completely. *where*
- Digital Terrestrial TV is a technology where allows you to receive more channels and a better picture through a conventional aerial instead of a satellite dish. *that*

37.3 Look at B and C opposite. Underline all the classifying expressions in this text.

Computer organization

A computer consists of hardware and software. The CPU, main memory and peripherals constitute what is known as hardware – the physical parts. The control unit, the arithmetic-logic unit and the registers are the basic parts of the CPU. The RAM and the ROM make up the main memory. Peripherals are classified into three types: input, output and storage devices.



Software can be divided into two categories: (i) system software, which includes operating systems, programming software and system utilities, and (ii) application software, which comprises programs that let you do specific tasks (e.g. graphics, email).

37.4 Complete the sentences with a classifying expression from the box.

there are two types of ...	... is composed of	... is a type of
... are made up of	there are four main classes of ...	

- microchips: (i) microprocessors, used as CPUs in computers, (ii) memory chips, used to store data, (iii) digital signal processors, used in mobiles and digital TVs, and (iv) application-specific integrated circuits, used in cars and appliances.
- In the future, people may have biochips inserted under their skin. Biochips two components: a small chip, called a transponder, and a scanner.
- A network two or more computers connected together to share information and resources.
- network architecture: peer-to-peer, where all PCs have the same capabilities, and client-server (e.g. the Internet), where servers store and distribute data, and clients access this data.
- Bluetooth wireless technology for transferring data between devices.

You and computers

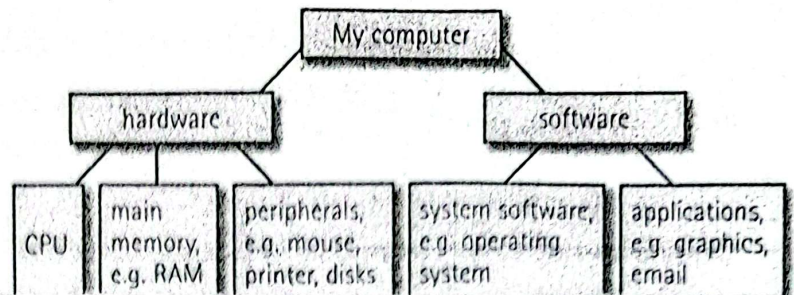
Draw a diagram classifying the hardware and software components of your PC. You can use exercise 37.3 to help you. Include as many details and devices as possible.

better picture through a conventional aerial instead of a satellite dish.

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*when choosing between the two
in general, only one option*

Which or what

- 1 Please let me know what time I should arrive.
- 2 You will find the office in West Street, which is in the 5th district.
- 3 Which of these two designs do you prefer?
- 4 I'm wondering what you hope to achieve in this meeting.
- 5 Do you know which of these two figures is correct?
- 6 The presentation will take place at the Hilton Hotel, which is a few minutes' walk from the station.

Make a relative sentence using a relative pronoun.

- 1 We are a new company ^{that} specializes in the production of semiconductors.
- 2 We were give your name by Mr Tom Furlan. ^{who} We believe ~~he~~ is one of your regular customers.
- 3 We enclose a complete product list. ^{where} We hope you will like ~~them~~.
- 4 We began selling our products six months ago at a local market. ^{where} ~~There~~, they proved to be very popular.
- 5 He works for a big multinational company. ^{whose} ~~Their~~ products are the top of the quality.

LANGUAGE WORK Relative clauses with a participle

Relative clauses with a participle are often used in technical descriptions. They allow you to provide a lot of information about a noun using as few words as possible.

Study these examples from the Task 3 text:

- 1 The technology *needed to set up a home network*
- 2 PCs *equipped with Ethernet adapters*
- 3 Network modem *allowing clients to access the Internet simultaneously*
- 4 Data line *linking client to server*

We can use the passive participle as in examples 1 and 2:

- 1 The technology *needed to set up a home network*
= technology *which is needed*
- 2 PCs *equipped with Ethernet adapters*
= PCs *which are equipped*

We can use an active participle as in examples 3 and 4:

- 3 Network modem *allowing clients to access the Internet simultaneously*
= modem *which allows clients to access the Internet simultaneously*
- 4 Data line *linking client to server*
= data line *which links client to server*

4 Complete these definitions with the correct participle of the verb given in brackets.

- 1 A *gateway* is an interface (*enable*) dissimilar networks to communicate.
- 2 A *bridge* is a hardware and software combination (*use*) to connect the same type of networks.
- 3 A *backbone* is a network transmission path (*handle*) major data traffic.
- 4 A *router* is a special computer (*direct*) messages when several networks are linked.
- 5 A *network* is a number of computers and peripherals (*link*) together.
- 6 A *LAN* is a network (*connect*) computers over a small distance such as within a company.
- 7 A *server* is a powerful computer (*store*) many programs (*share*) by all the clients in the network.
- 8 A *client* is a network computer (*use*) for accessing a service on a server.
- 9 A *thin client* is a simple computer (*comprise*) a processor and memory, display, keyboard, mouse and hard drives only.
- 10 A *hub* is an electronic device (*connect*) all the data cabling in a network.

37.1 Look at A opposite. Match the two halves to make correct definitions.

- | | |
|-----------------------------------|---|
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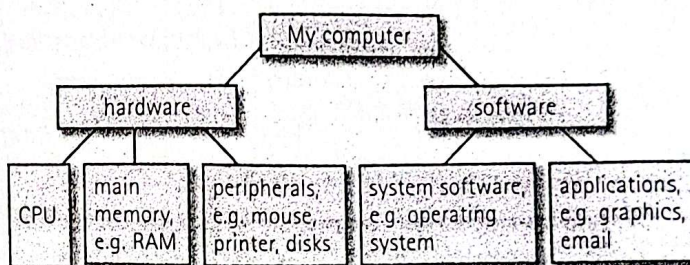
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37.4 Complete the sentences with a classifying expression from the box.

- | | | |
|------------------------------|--------------------------------------|--------------------|
| 1 there are two types of ... | 3... is composed of | 5 ... is a type of |
| 2 ... are made up of | 4 there are four main classes of ... | |

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- Bluetooth wireless technology for transferring data between devices.

You and computers

Draw a diagram classifying the hardware and software components of your PC. You can use exercise 37.3 to help you. Include as many details and devices as possible.

An overview of present-day satellite applications and services

Today, there are around 5000 satellites orbiting our Earth, of which over 2000 are operational today. According to the radius and the orbit period, most of these satellites can be classified into three types of orbits: the GEostationary Orbit (on a fixed orbital position in the sky around 36 000 km above the equator), the Mid Earth Orbit (between 2 000 km altitude and below GEO) and the Low Earth Orbit (typically less than 2 000 km above the earth's surface).

These three types of orbits roughly correspond to the three most important groups of satellite applications: television distribution and internet communications (GEO), position and time determination or navigation (MEO), and earth observation (LEO). There is a growing number of exceptions and many services are combinations of satellite applications.

1. Television signal distribution is still commercially the most important application of satellites. Television content is collected in **uplink ground stations (teleports)** and a high frequency signal is transmitted to a geostationary satellite according to the Digital Video Broadcast standards. The satellite retransmits the signal on a different frequency to earth so that millions of households can receive it by means of a small rooftop satellite dish and a satellite receiver. The transmitted content can also be redistributed on earth to a local **terrestrial** cable network.
2. Satellite News Gathering allows broadcasters to transmit live images from a news event to the broadcaster's studios. This service is particularly known from the SNG-vans that are often parked in the vicinity of television-covered news events.
3. Data Relay Satellites in Geostationary orbit, such as the American TDRS and the European EDRS satellites, can establish a temporary or continuous data link between a remote low orbit satellite or a remote ground station and the user's home base.
4. Internet connectivity can today be offered through satellites. This is most useful for people living in mountainous regions and remote islands, where **terrestrial coaxial or fibre optic cable** are too expensive, or even impossible to install. The O3B and the upcoming O3B mPower constellation can offer fast and **low-latency** ('delay') internet connections for the other 3 billion people on the globe who can't rely on terrestrial telecommunications infrastructure. Today, people travelling by train, ship or plane expect internet connectivity while they are on the move. This service can now be offered thanks to **steerable** antennas and satellite coverage over the oceans. Internet connectivity is the fastest growing commercial business offered by satellites today.
5. Machine-to-machine (M2M) and Internet-of-Things (IoT) are also internet connectivity applications. The targeted data stream is here often from a remote device to a controlling entity. This happens in energy production areas, such as windmill parks. In the future, we expect satellite communication can support road infrastructure and environmental sensors, including the broadcasting of update software to any type of connected device. Crew and passenger safety monitoring as well as **telemetry** on trains, ships and planes can also be supported by satellite communication. This can be M2M or M2Human communication.
6. GNSS stands for Global Navigation Satellite System. This is a world-wide system of satellites that allows a user to determine his time and position on earth. The most known

constellations are the American Global Positioning System GPS and the European Galileo, but there are also the Russian Glonass and the Chinese BeiDou. These satellites are typically MEO satellites at altitudes between 20000 and 24000 km above the earth. If you can receive at least 4 different satellite signals from any constellation with a small GNSS receiver, you can determine your position and time. Today's smartphones can receive such GNSS signals. This enables Location Based Services, such as local information distribution and location-based payments. Track & trace services determine the position and time of trucks, trailers and containers and transmit them to the owner's home base.

7. In some geographic areas, Satellite Based Augmentation systems like WAAS (USA), EGNOS (Europe) use GEO satellites to offer more measurement accuracy and integrity to the position determination process. This is very useful in aviation for GNSS supported landing.
8. Using differential GNSS or Real-Time Kinematic receivers, high precision position determination becomes achievable for agriculture, surveying, construction works, **dredging**, mining and oil extraction platforms. It becomes possible to document the exact position of gas pipes. Dikes and land masses can be observed continuously to predict and announce **dike breaches** and **land mass sliding**. Early Warning Systems can transmit warnings to the population.
9. Earth Observation satellites store pictures of the earth from a Low Earth Orbit and transmit them to a ground station, often located near the North or South Pole. Copernicus is the European programme that put in place the Sentinel satellites, successors of the Envisat and ERS observation satellites. A specific challenge for Low Earth Orbit satellite systems is the data repatriation from a remote ground station to the home base of the user. This can be done by means of data relay satellites, as mentioned in point 3. Earth Observation services are useful for meteorology, cartography, agriculture, environmental monitoring and monitoring in relation to disasters like forest fires and floods.
10. The Automatic Identification of Ships (AIS) system can detect and identify each vessel on earth that is equipped with an AIS transmitter. Its extension the VHF Data Exchange System (VDES) can send/receive **maritime** critical data to any equipped vessel.
11. Satellites can support ADS-B (Automatic Dependent Surveillance-Broadcast) for aviation by capturing the ADS-B signals in space and retransmitting the content to a ground station. In this way, no plane should still get lost above an ocean (such as happened to flight MH-370 in 2014).
12. All satellite applications and services (satellite communication, satellite navigation and earth observation) are useful in situations where the traditional terrestrial telecom infrastructure cannot be used because of destruction or congestion. This may be the case in areas touched by natural or man-made disasters, terrorist attacks and cyberattacks. It is also the case in war zones. Satellite communication systems like emergency.lu may be set-up in troubled areas to replace damaged mobile telephone infrastructure, support crisis centre operations, news gathering and telemedicine. Satellite navigation may be used for search and rescue operations and the tracing of humanitarian aid convoys. Earth observation may be used for disaster assessment including the determination of threatened areas.

probleme
naviga
pobavljane
informacija

13. Related to point 12 above, all satellite applications and services can be useful in military operations, at land, sea and air. Satellites can connect remote radio networks to military headquarters networks in a secure way.
14. Satellites can connect remote cellular base stations to a telecommunications backbone network, so that even the most remote regions can get mobile phone coverage. The corresponding services are called **cellular backhaul services**. Satellites will be an integral part of the upcoming 5G telecommunication networks.
15. Quantum Key Distribution services allow the transmission of random encryption keys without the possibility of any undetected interception by a third party. Satellites can generate random encryption keys and transmit these through optical satellite communication as photon states (qubits) to the end-user. The quantum properties of the photons ensure that any **eavesdropping** cannot pass undetected.
16. In the area of science, satellites can make observations for astronomy and/or support the operation of deep space communications to remote space probes.

In general, satellites play a role in countries in development and in remote areas. They often play a role in matters that affect our earth (climate, agriculture and environment).

Almost all satellite applications and services have required government funded research and development activities. Some of these services, like television distribution, internet connectivity and location based services, went commercial so that the operational services expenses can now be carried by commercial revenues. Other services are often related to government tasks such as ensuring the safety and security of citizens. This is the case for GNSS, including EGNOS services, and for crisis management through the Govsatcom programme. In these cases, the operational expenses need to be carried by the national governments. International collaboration, however, has allowed to **deploy** these satellite services on an international scale, so that important **economies of scale** could be achieved.

Answer the following questions:

General Understanding

1. How many satellites are currently operational out of the approximately 5000 orbiting Earth?
2. What are the three main types of satellite orbits, and at what altitudes do they typically operate?
3. How do the types of satellite orbits correspond to the main satellite applications?

Specific Applications

4. Explain how television signal distribution works via satellites.
5. What is Satellite News Gathering (SNG) and how is it typically used?
6. Why is satellite internet connectivity particularly useful in remote regions?

Technical Details

7. What is the difference between Machine-to-Machine (M2M) communication and Internet-of-Things (IoT) in the context of satellite applications?
8. What does GNSS stand for, and what are some of the key constellations within this system?
9. How do Satellite Based Augmentation Systems like WAAS and EGNOS improve GNSS services?

Earth Observation and Safety

10. What are some applications of Earth Observation satellites, and what challenges do they face?

Emergency and Military Applications

11. In what situations are satellite applications particularly useful when traditional terrestrial telecom infrastructure is unavailable?

Advanced and Future Applications

12. What are Quantum Key Distribution services, and how do satellites facilitate this technology?
13. How can satellites support scientific observations and deep space communications?

Commercial vs. Government-funded Services

14. Which satellite services have become commercially viable, and which still rely on government funding?
15. How has international collaboration contributed to the deployment of satellite services on a global scale?

Critical Thinking

16. Discuss the potential future developments in satellite technology and their possible impacts on society.
17. Evaluate the importance of government-funded research and development in the progression of satellite technologies.

space exploration
mapping the space
producing solar energy from space
(still under research)
autonomous driving
smaller satellites, that will be cost-effective
that would improve connectivity
and communication

issue: what to do with
inactive satellites
geopolitical issue - own
the satellites, data, tech

LANGUAGEWORK

Emphasising: cleft sentences

Compare these sentences:

- 1 We need an electronic version available anywhere and updated regularly.
- 2 What we need is *an electronic version available anywhere and updated regularly*.

In (2) the object of the sentence is made more important. We can use *What ... be* in this way to emphasise the subject or object of a sentence.

Now compare these:

- 3 Babbage invented the world's first mechanical computer.
- 4 It was *Babbage* that invented the world's first mechanical computer.
- 5 It was *the world's first mechanical computer* that Babbage invented.

We can use *It is/was ... that* to emphasise almost any part of a sentence.

4 Rewrite these sentences to emphasise the words in *italics*.

- 1 *It was in 1993* that Bill Gates introd.
- 2 Bill Gates introduced the Windows NT operating system *in 1993*.
- 3 *Bill Gates* introduced the Windows NT operating system in 1993.
- 4 Bill Gates introduced *the Windows NT operating system* in 1993.
- 5 I need *information* quickly. *What I need quickly* is information.
- 6 Ebooks will replace *all the throwaway books we read*.
- 7 *Ebooks will replace all the throwaway books we read*.
- 8 It's not *cheap*. *What it is not* is cheap.
- 9 I like *the look and the feel of books*. *What I like* is the look and the feel of books.
- 10 I don't like *reading off a computer screen*. *What I don't like* is reading off a screen.

6. ~~What will ebooks~~
It is all the thr. bo. that we read that the ebooks will replace.
7. It is the e-books that will replace.

5 With the help of these facts about the history of PCs, correct the statements which follow using any of the structures studied in this unit.

1968

Engelbart demonstrated first mouse.

1982

Acorn produced the BBC Micro.

1971

Intel designed first microprocessor.

1984

Apple Macintosh

1975

MITS Altair 8800 – home construction kit personal computer

1990

Windows 3.0

1975

Gates and Allen founded Microsoft in Albuquerque, New Mexico.

1993

Intel Pentium, Microsoft Windows NT

1980

Sinclair ZX80

2001

Apple launched OS X operating system. Microsoft launched Windows XP.

1981

IBM Personal Computer with Microsoft MS-DOS operating system

What Intel did was design first microprocessor.
It was designing first microprocessor that Intel did.

WORD STUDY Prefixes, -ise verbs

These prefixes are common in IT terminology.

Quantity

prefix	meaning	examples
bi	two	bi-directional
uni	one	uni-directional
tera	10^{12}	terabyte*
giga	10^9	gigahertz
mega	10^6	megahertz
kilo	10^3	kilohm
milli	10^{-3}	millisecond
macro	large	macroinstruction
micro	small	microcomputer
multi	many	multimedia

Prepositions

prefix	meaning	examples
anti	against	anti-virus software
infra	below	infrared
intra	within	intranet
inter	between	Internet
hyper	above, beyond	hyperlink
tele	distant, far	telecommunications

Others

prefix	meaning	examples
auto	by itself	automated
cyber	machine control	cyberspace
super	higher in quantity or degree	superhighway

(* approximate decimal value)

7 Find an IT term for each of these definitions. Each term includes one of the prefixes listed above. All of the terms are used in this book.

- working at a distance *teleworking*
- a system of numbers with 2 as its base *binary*
- a way of communicating between a user and a computer *interface*
- 1048576 bytes of information *megabytes*
- describes a program which allows two way communication between user and computer *interactive*
- set of computer instructions operating as one unit *macroprogram*
- having many different modes of input *multimodal*
- a computer higher in scale than any other *supercomputer*
- a machine which provides cash to bank customers without requiring a human operator: an A ... Teller Machine *automatic*
- a very small but powerful processor *microprocessor*
- a system used by many people *multiuser*
- a wordprocessing feature which corrects by itself *autocorrect*

17. Evaluate the progression of satellite ...

8 Verbs ending in *-ise* (US *-ize*) often have a causative meaning.

For example:

New phones will *revolutionise* the way we communicate.

= New phones will *make a revolution in* the way we communicate.

Replace the words in *italics* in each sentence with the appropriate form of an *-ise* verb. All the verbs have been used in this book.

- 1 Players let you group songs into playlists and *make the selection random*. *randomise the selection*
- 2 If you adopt differential backup, this *reduces to a minimum* the size of your backup set. *minimises*
- 3 Most hotels use *systems which have been converted for computers*. *computerised*
- 4 Software developers can produce *solutions which are tailored to the customer*. *customised*
- 5 Some software houses produce *specially-written* applications. *specialised*
- 6 Utilities can be *put into categories* as editors, filters or communications programs. *categorised*
- 7 You can protect data by putting it in a form only *users with authority* can understand. *authorised*
- 8 It is an offence to make *copies of software which are done without authority*. *unauthorised*

SPEAKING

9 What features should an electronic book reader have? In your group decide on the best specifications. Compare your list with the reader described on the recording and, if possible, with the specifications for the readers described at these websites:

www.franklin.com
www.ebook-gemstar.com
www.bookstore.glassbook.com

10 Now present your specifications to the rest of the class. Each member of your group should present some of the specifications with reasons for their choice.